

# Survival Model Evaluation: `chicago_licences`

*Fitted with the strategy-survival-model pipeline; methodology validated separately against synthetic ground truth*

|                   |                                                                                          |
|-------------------|------------------------------------------------------------------------------------------|
| Source data       | <code>datasets/chicago_licences.csv.gz</code>                                            |
| Rows              | 262,763 (83.6% with the ending observed, 16.4% censored)                                 |
| Start dates       | 2002-01-02 to 2026-07-31                                                                 |
| Evaluation        | 5 expanding-window temporal folds                                                        |
| Source of figures | <code>reports/chicago_demo/metrics.json</code> , regenerated by the command in section 7 |

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## 1. Summary

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This report evaluates a survival model fitted to `chicago_licences.csv.gz`, 262,763 rows, each observed from its start date for 'licensed\_days' days with 'closed' marking whether the ending was seen (83.6%) or the row was still running when observation stopped (16.4%). Median observed duration is 759 days. The model predicts, from the 300 features available at the start date, how long each row survives.

Pooled over 157,658 out-of-time test rows, the XGBoost AFT model reaches a concordance index, the share of pairs it puts in the right order, of 0.697 (95% bootstrap interval 0.695 to 0.699), against 0.500 for a coin flip. Set beside a Cox proportional hazards baseline on the same features it scores 0.695 to the baseline's 0.695; both of those are fold means, which is the only like-for-like basis and is why the pooled figure is not the one compared. Most of the pooled figure reflects a row's `license_description` group rather than ranking within it. Ranking every row by its group's mean prediction alone scores 0.703, and comparisons within a group score 0.544; the results section gives the decomposition. At the 1095-day horizon its censoring-weighted Brier score is 0.206, beating the 0.247 of a no-skill forecast that assigns every row the population average.

That bootstrap interval is narrow, and it is worth saying what it does and does not cover. It holds each fold's fitted model fixed and resamples the test rows, so it measures how precisely this run's score is pinned down by the number of rows scored, nothing more. It says nothing about how much the score would move on a different stretch of history, and the per-fold spread of 0.604 to 0.782 in Table 2 is the better guide to that. The rows are also unlikely to be independent, since rows that share a category or a stretch of time tend to fail together, which makes the true interval wider than the one printed.

This dataset has no known generating process. Unlike the synthetic validation report, there is no oracle ceiling to say how much signal remains unclaimed, and feature attributions cannot be checked against a true mechanism. What carries over from the synthetic validation is the pipeline itself: the same temporal folds, label re-censoring, likelihood-based selection, and calibration checks, verified there against known answers.

## 2. Data

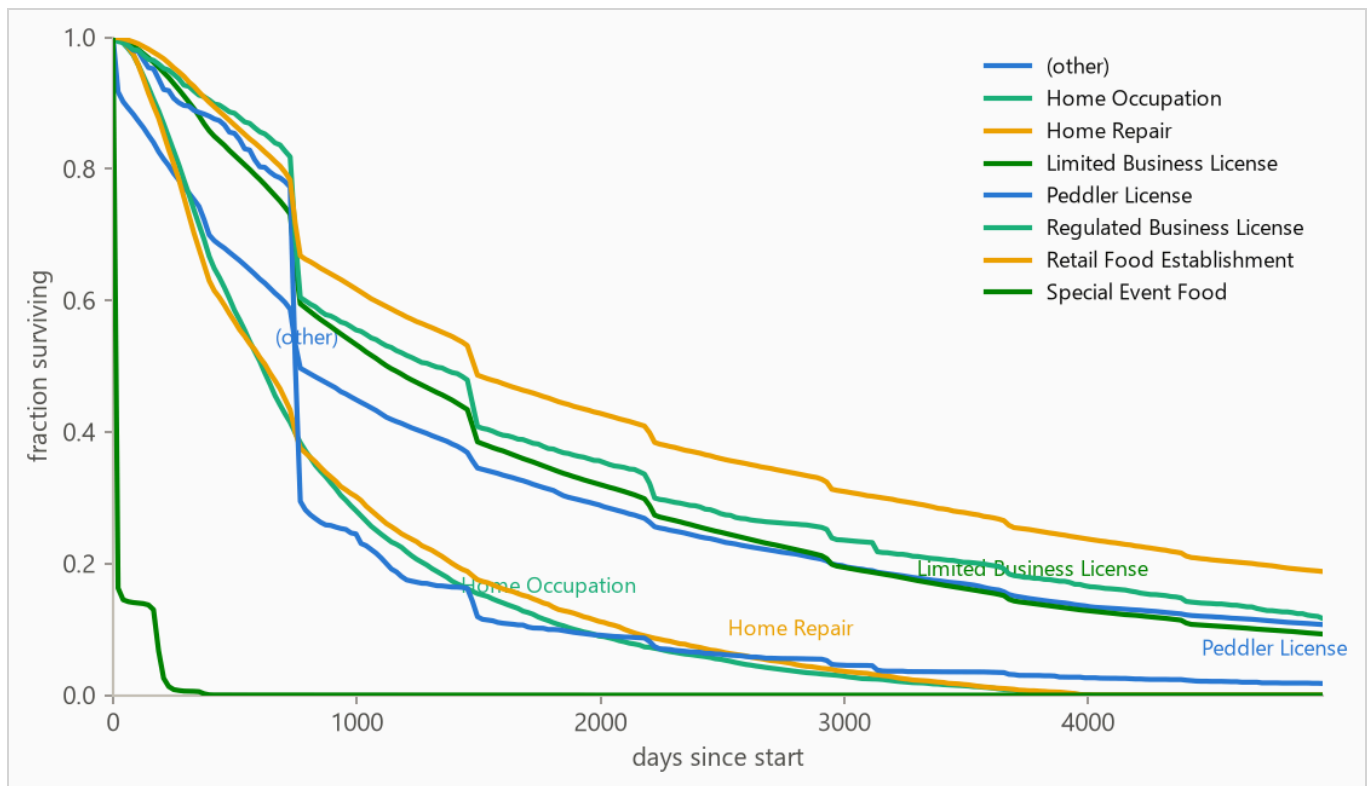
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The dataset is `datasets/chicago_licences.csv.gz` : 262,763 rows, each observed from its start date. Start dates run 2002-01-02 to 2026-07-31. 83.6% of rows have their ending observed and 16.4% are censored, meaning the row was still open when observation stopped, so its full duration is unknown; median observed duration is 759 days. Text columns are one-hot encoded via the saved encoding recipe: `ward`, `community_area`, `police_district`, `zip_code`.

The pipeline treats every row as observed from its own start date. It has no support for left truncation, rows whose life began before the source window opened and that would enter the data mid-life, so any such rows have to be excluded when the dataset is prepared. Whether they were, and how many, is a property of the preparation step recorded with the dataset.

For this dataset the preparation step did perform that exclusion. Licences already running when the portal's history begins enter at their first recorded renewal rather than their true start, and 61,351 of them, 23 percent of the pull, were removed by keeping only licences whose earliest transaction is a genuine issue. The rule and the counts are recorded in `scripts/run_prepare_chicago.py`.

Figure 1 shows Kaplan-Meier survival curves by `license_description`, the coarsest structure in the outcome before any model is fitted. A Kaplan-Meier curve estimates the fraction of rows still surviving at each age, with censored rows counted for as long as they were observed.



**Figure 1.** Kaplan-Meier survival curves by `license_description`. Groups beyond the seven most frequent are collapsed into (other) for this plot only. The vertical drop in the (other) curve at about two years, and the smaller steps at term multiples in other curves, are licence terms expiring. A business that does not renew exits at a term boundary, so endings cluster there.

## 3. Method

### 3.1 Model class

The target is a duration with incomplete observations, so the model is an accelerated failure time (AFT) model, implemented as XGBoost with the `survival:aft` objective. Censoring enters through interval labels, where an observed death is the interval  $[t, t]$  and a censored row is  $[t, \text{infinity})$ .

Regression on lifetime was rejected because censored rows have no target, and dropping them discards the longest-lived rows, which biases the model toward pessimism. Classification at a fixed horizon handles censoring awkwardly and answers only one question. AFT keeps every row and returns a full time distribution, which collapses to any horizon of interest.

Concretely, the fitted model predicts one number per row, the median survival time in days. The full curve comes from wrapping a log-normal distribution of fitted width around that median, so the probability of

surviving any horizon is a point read off that curve. The width is a single value shared by every row; how it is selected and calibrated is covered in section 3.3.

Numeric features pass through as-is, missing values included, which XGBoost handles natively; the Cox baseline receives train-window median imputation. Text columns are one-hot encoded.

### 3.2 Temporal validation and label re-censoring

Rows are ordered by start date and evaluated with 5 expanding-window folds. The earliest 40% of rows is burn-in that is only ever trained on. Every fold trains only on rows that started before its split date and tests on the next block, so training sets grow from 105,103 to 231,220 rows.

Training labels are re-censored at each split date. A row started two years before a split may have died three months after it, and its recorded label contains that future death. Any model trained at that split date could only have known the row was still running, so every post-split death is rewritten as a censoring at the split. Omitting this step raises measured scores while silently importing future information, which makes it the most consequential detail in the pipeline. It is implemented as a standalone function ( `cv.recensor` ) with dedicated tests, because it transfers unchanged to any duration problem on operational data.

Test labels use the full observation window, because the restriction exists to keep the future away from the model, not away from the scorer.

### 3.3 Hyperparameter selection and calibration

Hyperparameters are selected once, on the first fold's training window, using an inner temporal split and held-out censored log-likelihood. Likelihood rather than concordance drives selection, and the reason is a failure this pipeline recorded rather than avoided.

An earlier version selected on concordance and looked correct, with ranking metrics landing where they ultimately did. It then lost to a no-skill reference forecast on 365-day Brier score, the mean squared error of a probability forecast. A concordance index is invariant to the predictive scale, so the search had no reason to prefer a usable distribution width and settled on one far too wide. Nothing in the training loop objected, because nothing in it measured the width. Selection now uses likelihood, which penalizes a broken scale, and the predictive log-normal scale is calibrated on a temporal tail slice by a probe model that never trained on those rows; the final model is then refit on the full window, including that slice, and carries the probe-calibrated scale. The selected loss scale was 0.9, and the calibrated predictive scale came out at 1.28.

The general lesson holds beyond this project. A model can rank well and misstate probabilities at the same time, and the only defense is evaluating both.

### 3.4 Baselines

A Cox proportional hazards model fitted on the same features is the standard survival alternative and answers whether the gradient-boosted model was necessary.

## 4. Results

### 4.1 Discrimination

Pooled over 157,658 out-of-fold rows, with 95% percentile bootstrap intervals over 500 resamples:

**Table 1.** Concordance index by method (Harrell's C is its standard estimator). Higher is better; 0.500 is a coin flip. Cox appears as a fold mean only, because its per-fold risk scores share no scale to pool.

| METHOD                               | CONCORDANCE (HARRELL'S C) | 95% INTERVAL   |
|--------------------------------------|---------------------------|----------------|
| XGBoost AFT (pooled)                 | 0.697                     | 0.695 to 0.699 |
| XGBoost AFT (fold mean)              | 0.695                     | not resampled  |
| Cox proportional hazards (fold mean) | 0.695                     | not resampled  |

On this dataset the Cox baseline scores 0.6951 against the boosted model's 0.6946 on fold-mean concordance, a margin that sets the default for scoring new rows without proving a real difference. Both fitted models are saved and the Cox model is the one this run recommends. Reporting the near-tie, rather than presenting the boosted model as the result, is the point of carrying a baseline at all. Each fold refits the Cox model, and its risk scores are relative ones centred on that fold's own training window, so they are meaningful within a fold but carry no common scale across folds; the fold-mean rows are the like-for-like comparison and the pooled figure is not comparable to them.

The per-fold spread deserves as much attention as the mean. The weakest window is fold 2, starting 2012-08-02, at 0.604, and no cause is established for it in this report. Any use of the pooled figure should carry that range: there are stretches of history where the model ranks these rows substantially worse than its headline number suggests.

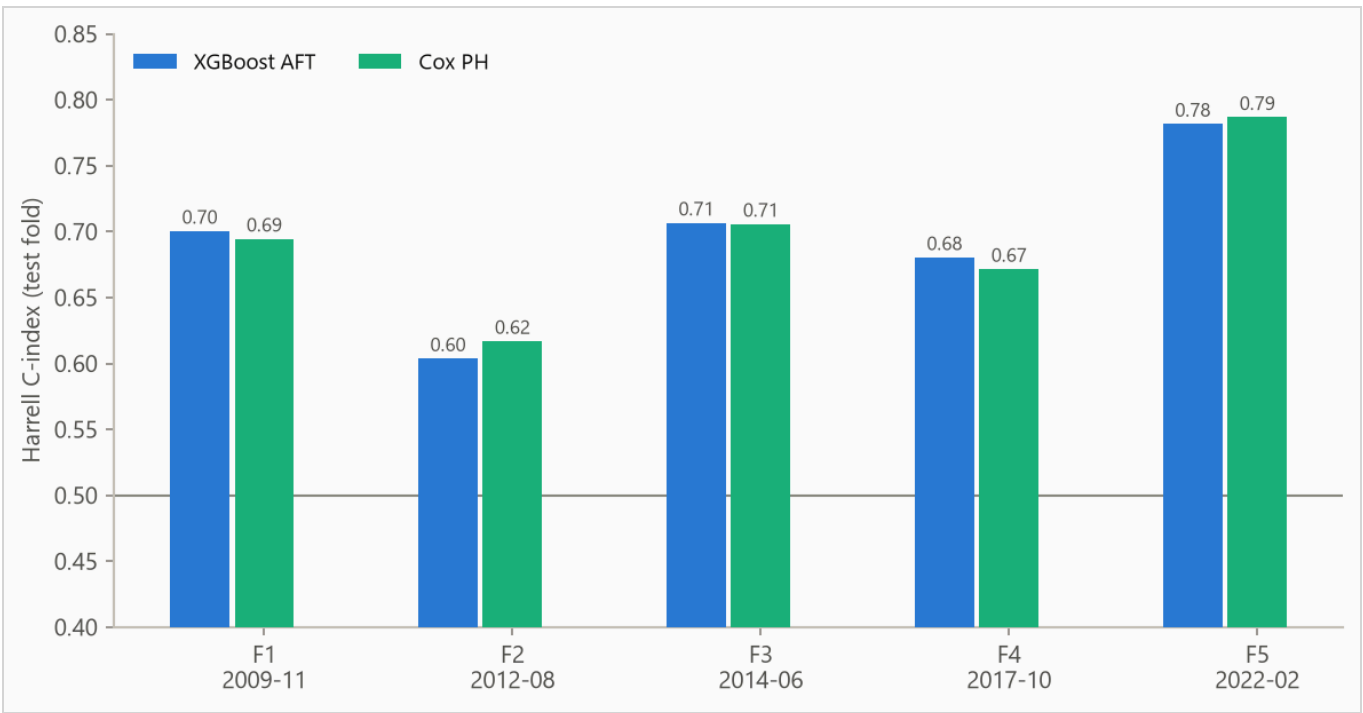
The pooled concordance also decomposes by `license_description`. Ranking every row by its group's mean prediction alone scores 0.703, and comparisons restricted to rows in the same group score 0.544, pair-weighted across 69 groups with at least 50 rows and 10 observed endings. The distance of the within-group figure above 0.500 is the row-level skill; the rest of the pooled figure is group membership. Both numbers come from the same out-of-fold predictions as the pooled figure. Stated plainly, the group means alone

outscore the full model's pooled figure; the within-group variation adds ranking skill inside a group and costs a little ordering across groups.

**Table 2.** Per-fold results. Censoring is the share of each fold's test rows whose ending was not observed.

| FOLD | SPLIT DATE | TRAIN N | TEST N | CENSORED | AFT   | COX   |
|------|------------|---------|--------|----------|-------|-------|
| 1    | 2009-11-09 | 105,103 | 31,532 | 6%       | 0.701 | 0.695 |
| 2    | 2012-08-02 | 136,592 | 31,532 | 14%      | 0.604 | 0.617 |
| 3    | 2014-06-09 | 168,159 | 31,532 | 16%      | 0.707 | 0.706 |
| 4    | 2017-10-23 | 199,693 | 31,531 | 27%      | 0.680 | 0.672 |
| 5    | 2022-02-23 | 231,220 | 31,531 | 60%      | 0.782 | 0.787 |

Figure 2 plots the per-fold concordance from Table 2; each fold's censoring mix is in the table.



**Figure 2.** Concordance by fold, from 0.604 to 0.782. Folds differ in censoring mix, so cross-fold comparisons are indicative rather than exact.

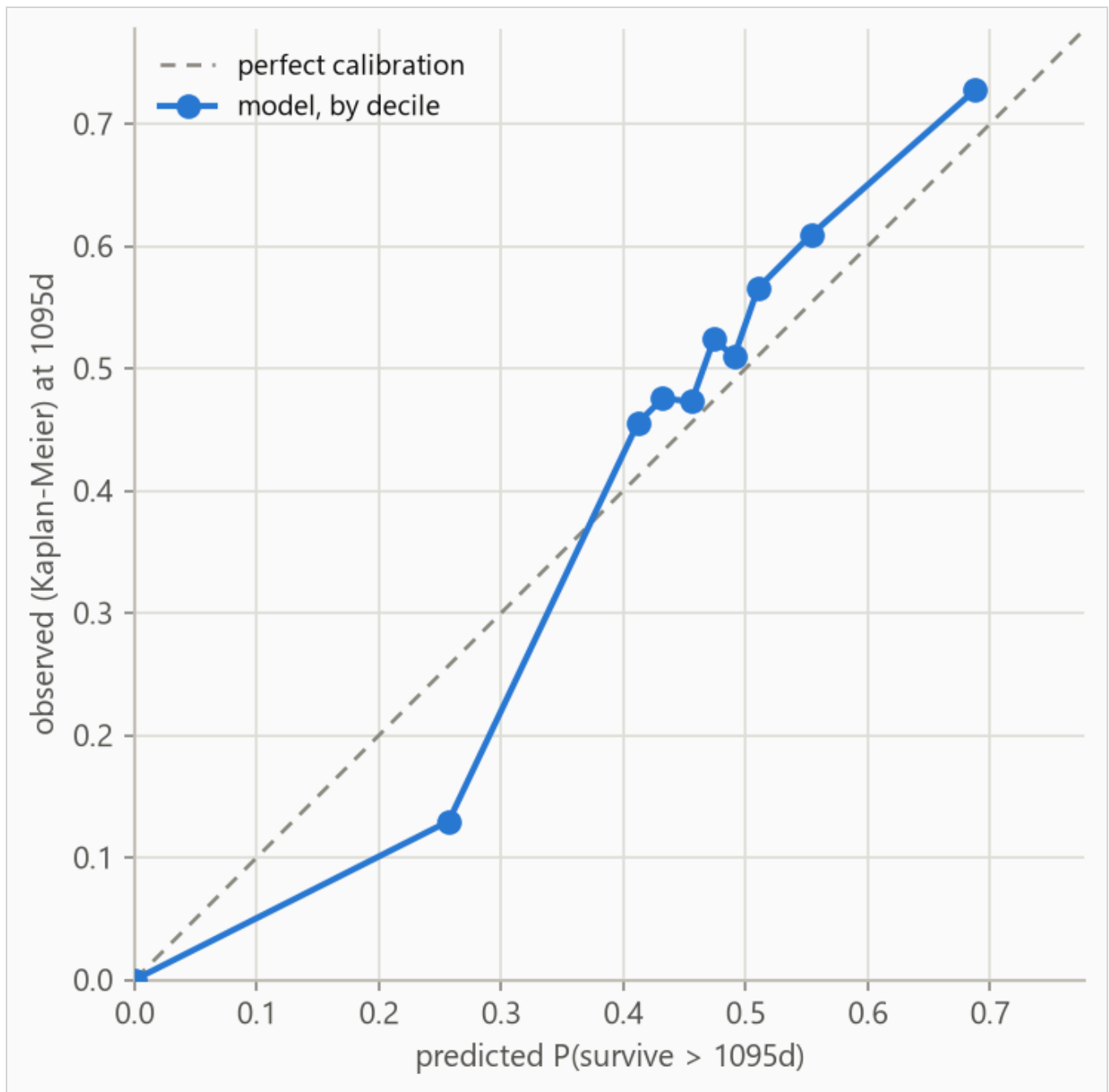
4.2 Calibration

Predicted median survival times are converted to survival probabilities under the calibrated log-normal, then checked two ways. The Brier score is the mean squared error of a probability forecast, lower being better, and here it is weighted by the inverse probability of censoring (IPCW), so that rows whose outcome was censored

away do not bias the result. The no-skill reference assigns every row the pooled Kaplan-Meier marginal, ignoring all features.

**Table 3.** IPCW Brier score by horizon. Lower is better.

| HORIZON   | AFT   | COX   | NO-SKILL MARGINAL |
|-----------|-------|-------|-------------------|
| 365 days  | 0.084 | 0.082 | 0.158             |
| 1095 days | 0.206 | 0.207 | 0.247             |
| 1825 days | 0.194 | 0.192 | 0.215             |



**Figure 3.** Predicted against observed survival at 1095 days, by predicted decile. Observed frequencies are Kaplan-Meier estimates within each bin, so censored rows contribute correctly. The largest deviation is 0.127 in decile 2, and it is a real miss, not display noise.



**Table 4.** Decile calibration at 1095 days, the values plotted in Figure 3.

| DECILE | N     | PREDICTED | OBSERVED (KM) | DEVIATION |
|--------|-------|-----------|---------------|-----------|
| 1      | 15797 | 0.000     | 0.000         | -0.000    |
| 2      | 15748 | 0.257     | 0.130         | -0.127    |
| 3      | 16166 | 0.412     | 0.455         | +0.044    |
| 4      | 15480 | 0.432     | 0.475         | +0.043    |
| 5      | 17372 | 0.456     | 0.473         | +0.017    |
| 6      | 14036 | 0.474     | 0.524         | +0.050    |
| 7      | 15761 | 0.490     | 0.509         | +0.019    |
| 8      | 16054 | 0.510     | 0.566         | +0.055    |
| 9      | 15601 | 0.553     | 0.609         | +0.056    |
| 10     | 15643 | 0.688     | 0.728         | +0.040    |

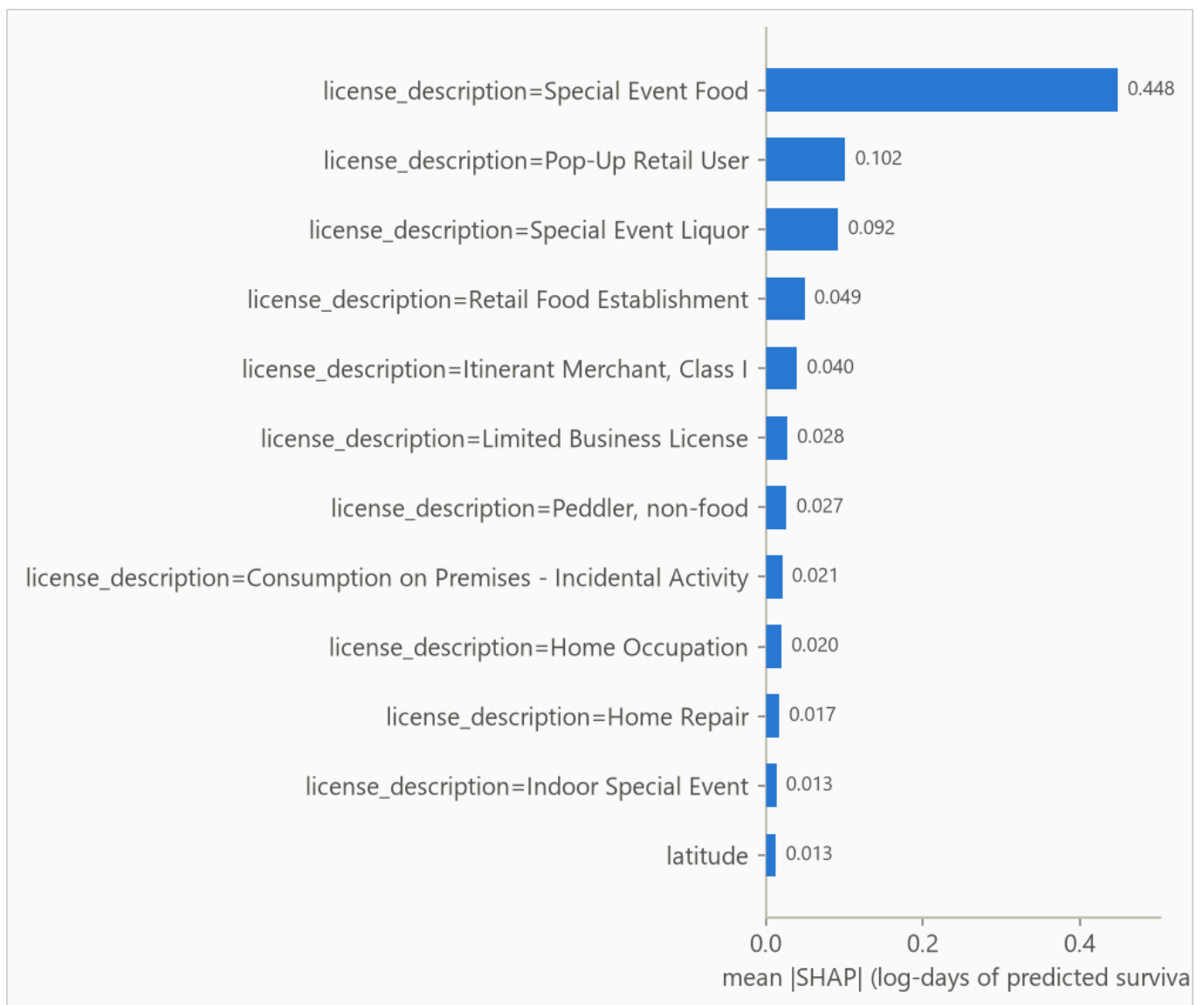
One caveat on the decile table. The observed value inside each decile is a Kaplan-Meier estimate, and when a decile's last observed row falls short of the 1095-day horizon the estimate carries its last value forward. In small or heavily censored deciles the observed column is softer than its three decimals look, and deviations there should be read accordingly.

## 5. What the model uses

Feature attributions (SHAP values) are computed on the log-time margin, so a value of +0.3 multiplies predicted survival time by about 1.35 and negative values shorten it.

On this data the attributions are descriptive only. There is no known mechanism to check them against, and correlated features can split credit in ways that reflect the model's internal choices as much as the data, so read the ranking as "what this model leaned on", not as importance in the world. They are also computed on the final model's own training rows, an in-sample reading rather than out-of-fold evidence.

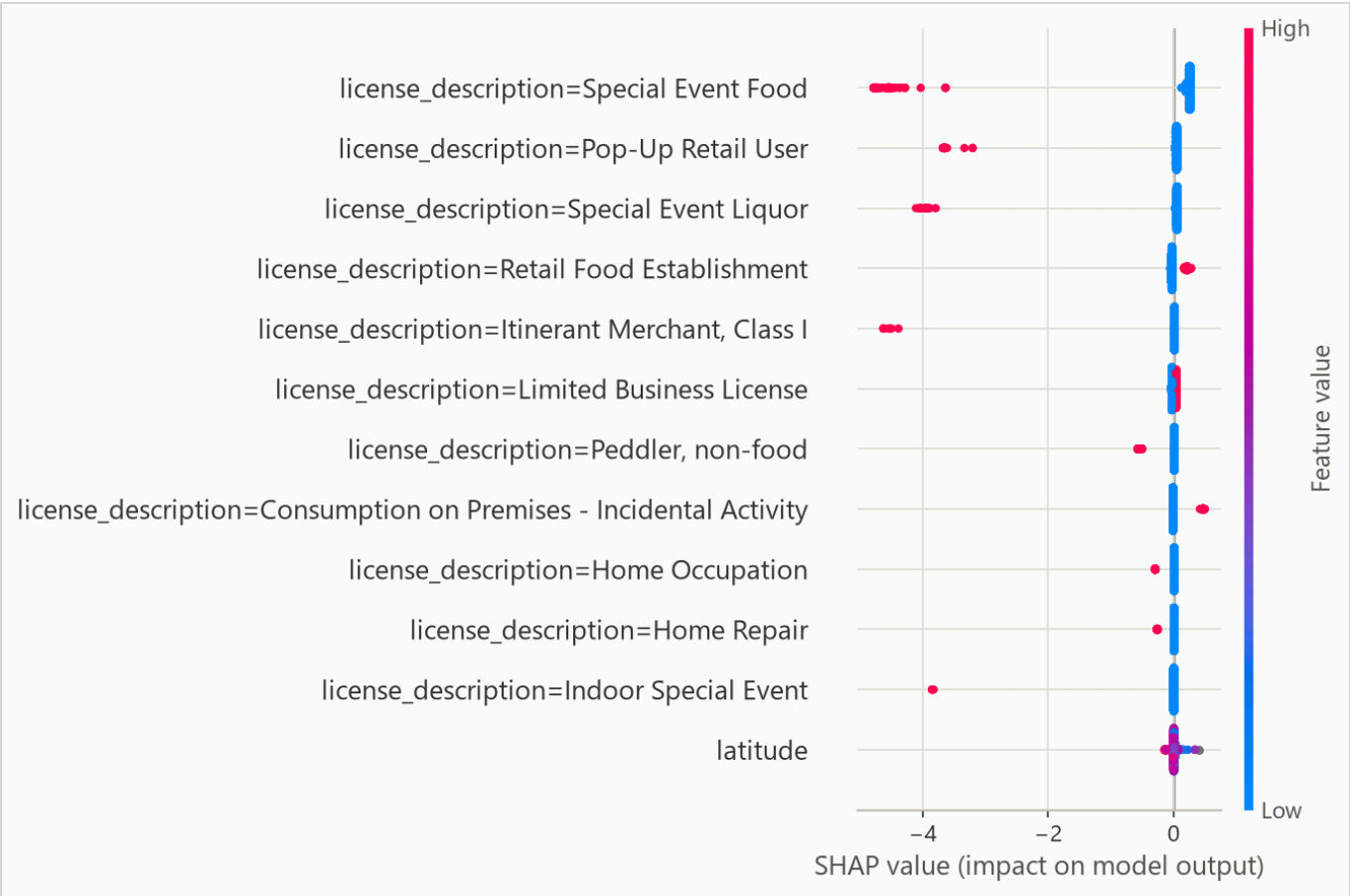
Figure 4 ranks features by mean absolute attribution, Table 5 lists the top eight, Figure 5 shows the per-row spread, and Figure 6 traces attribution against feature value.



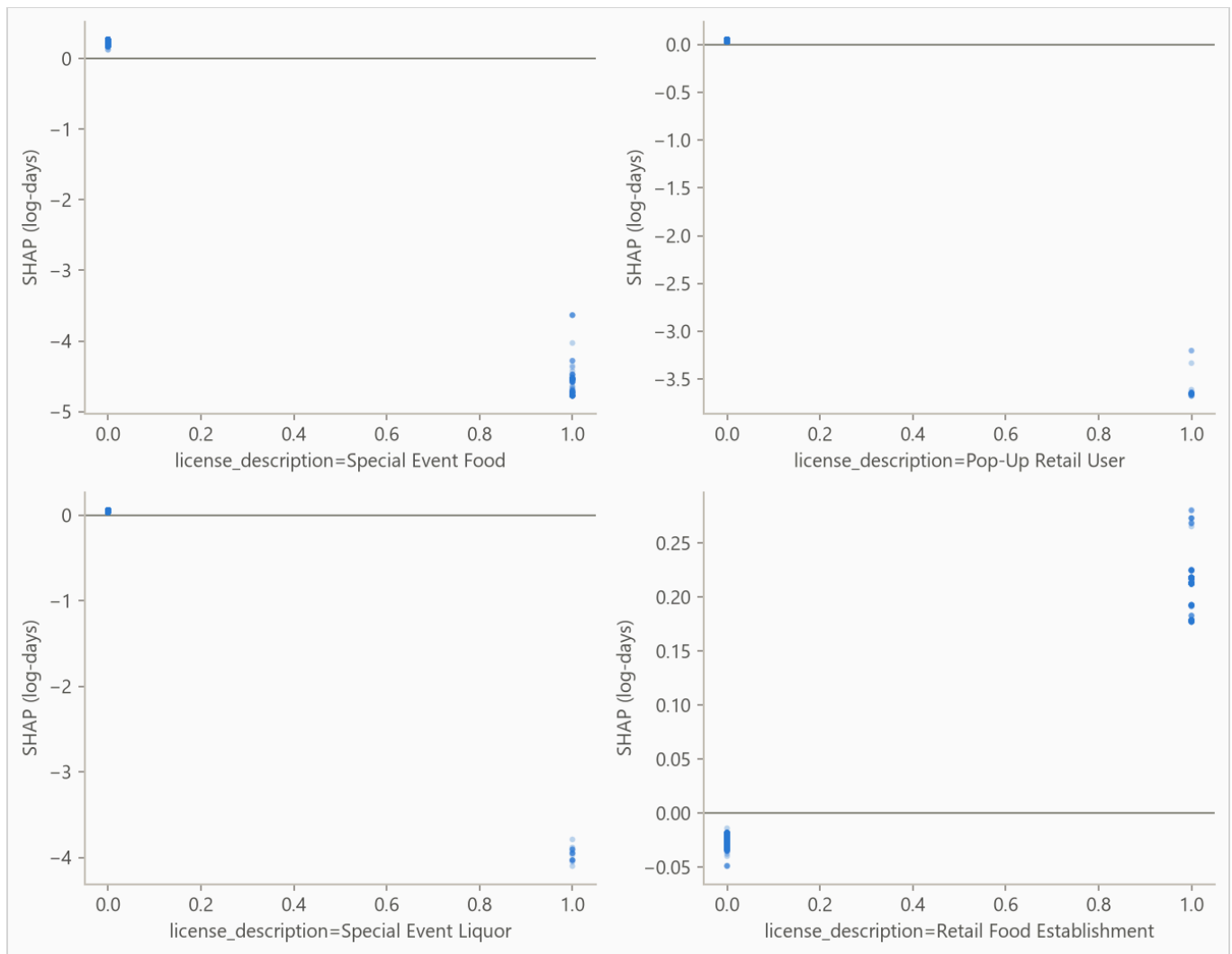
**Figure 4.** Mean absolute attribution by feature.

**Table 5.** Top eight features by mean absolute attribution.

| FEATURE                                                           | MEAN  ATTRIBUTION |
|-------------------------------------------------------------------|-------------------|
| license_description=Special Event Food                            | 0.448             |
| license_description=Pop-Up Retail User                            | 0.102             |
| license_description=Special Event Liquor                          | 0.092             |
| license_description=Retail Food Establishment                     | 0.049             |
| license_description=Itinerant Merchant, Class I                   | 0.040             |
| license_description=Limited Business License                      | 0.028             |
| license_description=Peddler, non-food                             | 0.027             |
| license_description=Consumption on Premises - Incidental Activity | 0.021             |



**Figure 5.** Per-row attributions across the explanation sample.



**Figure 6.** Attribution against feature value for the four features with the largest mean attribution.

## 6. Limitations

**No ground truth.** Nothing bounds how much predictable signal this dataset contains, so the concordance above cannot be placed relative to a ceiling the way the synthetic validation could.

**Censoring may be informative.** The evaluation assumes rows stop being observed for reasons unrelated to their risk. If observation ended early on rows that were about to end anyway, survival estimates are biased upward. Whether that holds here depends on how this dataset was collected.

**Endings near the cutoff are provisional.** Where no cancellation exists, a licence's end is its latest recorded expiry, so a licence whose term ran out shortly before the cutoff but that renews late is recorded as an observed closure. Deaths dated close to the cutoff therefore include some licences that will turn out to have survived; how many is not measured here.

**The encoding vocabulary is fitted on the full file.** One-hot columns and the collapsing of rare levels are decided from every row before the temporal folds are cut. No outcome labels are involved, but the early folds' feature matrices reflect level frequencies that were not knowable at their split dates.

**Every row shares one curve shape.** The predictive scale is a single fitted number, so the model shifts each row's survival curve earlier or later but never widens or narrows it, and two rows with the same predicted median receive identical probabilities at every horizon. Letting the width vary by row would need a model that predicts it, which is future work.

**Harrell's concordance is biased under heavy censoring.** The final fold is 60% censored, where an inverse-probability weighted concordance would be preferable. Fold-to-fold comparisons in Table 2 should be read with that in mind.

## 7. Reproducing this run

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```
python scripts/run_fit_evaluate.py --data datasets/chicago_licences.csv.gz --name chicago_licences
python scripts/run_build_report.py --run reports/chicago_demo
```

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Generated from `reports/chicago_demo/metrics.json` by `scripts/run_build_report.py --run .` 262,763 rows, 157,658 out-of-time test rows.